Understanding dissociation dynamics of CO_2 on Cu(110) surface and developing techniques for state selected molecule-surface scattering experiments

PhD Thesis Synopsis

by

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Introduction

Chemical reactions at a microscopic level are governed by atomic/molecular motion and their energetics. Studying the dynamic processes involved in chemical reactions on surfaces provides valuable insights into various aspects such as energy transfer pathways, timescales, reaction probabilities, and elementary processes like adsorption, diffusion, dissociation etc., elucidating atomic-level details.

To conduct detailed investigations of events at the gas-surface interface, it is essential to independently control and measure the translational, electronic, vibrational, and rotational levels of a molecule before and after collision with the surface. Molecular beam techniques make such experiments feasible. The primary motivation for studying these interactions lies in their direct relevance to heterogeneous catalysis processes [1]. Molecular beam surface scattering experiments involving nearly monoenergetic supersonic molecular beams with well-defined internal state distributions and interacting with well-characterized single-crystal surfaces provide a close approximation to the ideal scenario. Techniques such as time-of-flight (TOF) measurements and state-selective detection methods like resonance-enhanced multiphoton ionization (REMPI) are employed in these experiments. Surface-molecule interactions are broadly categorized into reactive scattering, involving chemical transformations, and non-reactive scattering, which entails only energy exchange without any chemical change. In reactive scattering, dissociative chemisorption is a process where a molecule chemisorbs on the vacant sites of the surface by breaking bonds also known as direct dissociation. This process requires the incident reactant molecule to have sufficient energy to overcome the activation barrier (see Figure 1, left) [2]. Another mechanism is precursor-mediated chemisorption also known as non activated dissociation pathway, where a molecule initially undergoes physisorption on the surface, gets trapped, and then finds the active site through thermal diffusion, ultimately leading to chemisorption (see Figure 1, right) [3]. In the context of non-reactive scattering, the incident molecules either (i) stick to the surface get thermally equilibrated, and then eventually get desorbed from the surface (i.e. the trapping desorption), or (ii) immediately leave the surface after collision (i.e. Direct scattering).

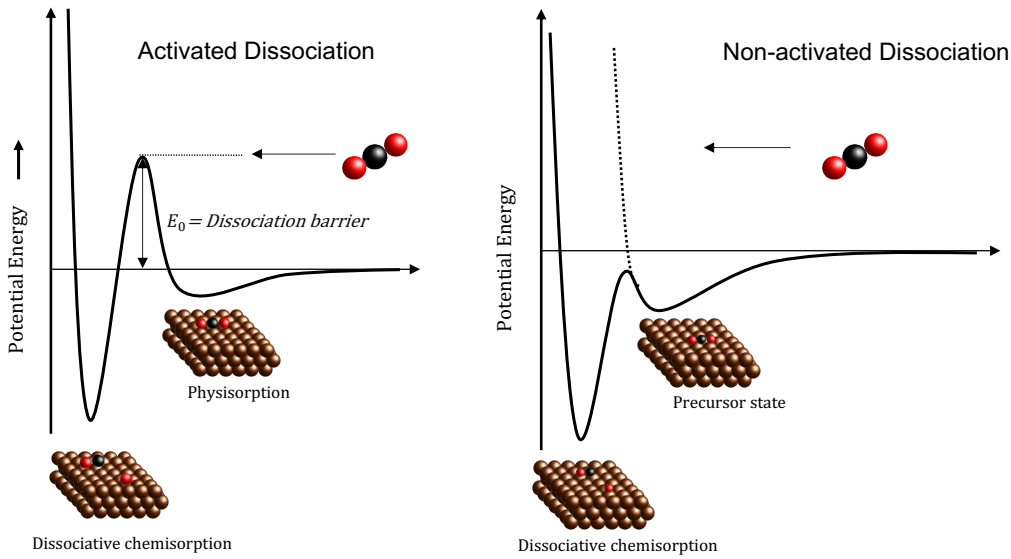

Figure 1: (left). A schematic 1D representation of molecule–surface interaction energy for direct dissociation process. (right) shows a schematic one-dimensional representation of potentials for precursor-mediated dissociation pathway.

Exploring the role of additional degrees of freedom, such as vibrational energy in

molecules-surface interactions, energy transfer pathways (surface phonon excitation or electron-hole pair excitation) reveals the nature of coupling among the molecular and surface degrees of freedom either adiabatic or non-adiabatic in non-reactive scattering [4, 5]. In the case of reactive scattering, it offers insights into the nature of potential energy surface, distinguishing between early and late barriers [6]. A comprehensive understanding of both reactive and non-reactive scattering, involving ground state and vibrationally excited molecules on metal surfaces, is crucial for testing and benchmarking outcomes of theoretical and computational models.

The primary objective of my thesis work is to comprehend the ground and mode-specific dissociation and scattering of CO_2 molecules at Cu(110) surface. To achieve this goal, I have actively participated in constructing and designing diagnostic tools that will facilitate the understanding of both reactive and non-reactive scattering of molecules at metal surfaces. In the initial phase of my thesis work, I was engaged in the design, testing, and characterization of a spatial imaging technique. This technique proves to be valuable for comprehending nonreactive scattering phenomena. Furthermore, I have designed and developed a high-precision wavelength meter which is a valuable tool for quantum state-specific studies. This device played a crucial role in identifying precise transition wavelengths and concurrently active wavelength stabilization of the home-built IR OPO, as needed in the molecular beam experements.

The overall document will follow the structure outlined below.

- Molecular beam surface scattering setup and its features
- Understanding of CO_2 dissociation dynamics on Cu(110) surface
- Design and characterization of Ion imaging tools for non-reactive scattering studies
- Ongoing work: Understanding of the role of vibrational energy in CO_2 dissociation dynamics on Cu(110) surface
- A novel approach to build high accuracy and precision wavemeter for quantum state specific surface scattering studies

1 Molecular beam scattering setup and its features

Below we provide a brief description of the salient features of our molecular beam scattering apparatus used to study gas-surface dynamics.

1.1 Experimental setup

The overall experimental setup for the surface scattering experiment is shown in Figure 2. It consists of a source chamber and two differential pumping stages (Diff-1 and Diff-2) followed by a UHV chamber where the target single crystal was placed. It is housed in a turbomolecular pumped vacuum chamber with a base pressure below (6 to 8) $\times 10^{-10}$ mbar, monitored using a nude ion gauge (IMR 430, Pfeiffer vacuum). The chamber is also equipped with a cylindrical mirror analyzer for Auger electron spectroscopy (AES) and a quadrupole mass spectrometer for thermal desorption spectrometry (TDS). The Cu(110) crystal can be resistively heated to 1200 K and cooled to less than 90 K. A K-type thermocouple is inserted into the side of the crystal for temperature measurements. The Cu crystal was initially cleaned by resistive heating and ion bombardment cycles, followed by annealing at 800 K. Surface cleanliness was confirmed by the absence ($\leq$ 1 %) of C(KLL), O(KLL), etc., in the AES. A pulsed solenoid valve with an opening diameter of 1 mm (Parker 009-1643-900, driver Iota One 060-0001-900) placed in the

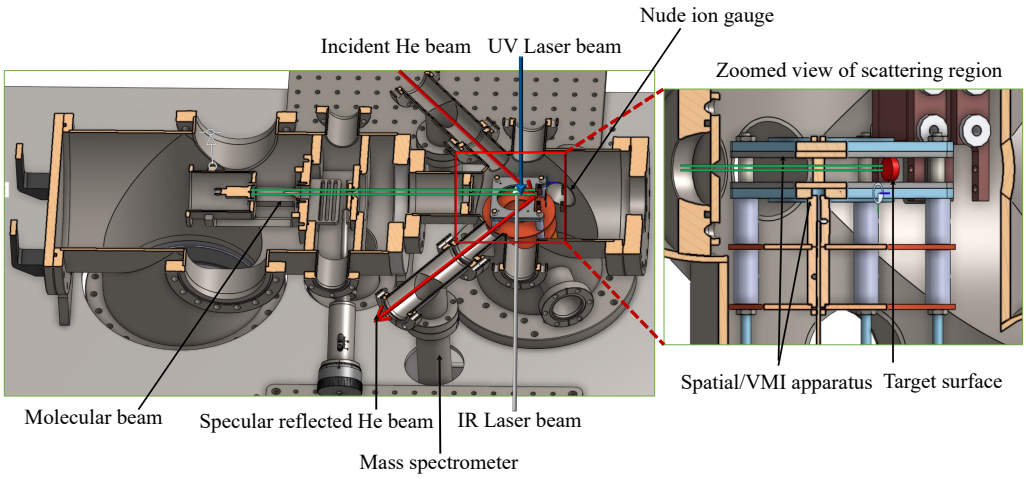

Figure 2: A cross-sectional drawing of the molecule surface of the scattering chamber setup. A close-up view of the rectangular red box highlights the spatial imaging/velocity map imaging setup, enabling the measurement of the incident and scattered flux velocity as well as angular distribution.

source chamber was used as a molecular beam source. The supersonically expanded gas was made to pass through a 1.5 mm opening diameter skimmer (Beam Dynamics) and two subsequent apertures (2 mm diameter) placed downstream at the entrance of Diff-2 and UHV chambers. Additionally, a mass spectrometer (SRS RGA 200), calibrated with the ion gauge as a reference, was used to measure the residual gas composition and estimate the incident beam flux. The CO_2 flux on the target surface ranged from 0.05 - 0.8 ML/sec, where 1 ML corresponds to 1.08×10^{15} atoms cm^{-2}, the surface density of Cu atoms on a flat Cu(110) surface.

1.2 Molecular beam preparation

A molecular beam is a well-collimated stream of electrically neutral molecules traveling in a collision-free environment, essential for studying collision and reaction dynamics at metal surfaces. It is produced by expanding gas through an orifice into a low-pressure region, molecular beams can be effusive ($\lambda >$ d) or supersonic ($\lambda \ll$ d), depending on expansion conditions (stagnation pressure P_0 and temperature, T_N), where λ is the mean free path and d is orifice diameter. In supersonic expansion, thermal molecular motion is converted into directed motion, yielding a nearly monoenergetic beam with a terminal velocity ($\bar{V}_\infty$).

$$\bar{V}_\infty = \sqrt{\frac{2\bar{C}_p}{\bar{M}} T_N} \tag{1}$$

The terminal velocity is determined by molar average heat capacity ($\bar{C}_p$), molar average molecular weight ($\bar{M}$), and temperature (T_N) at the nozzle. By adjusting expansion parameters, such as P_0, $\bar{C}_p$, and T_N, we can control the incident energy of the molecular beam, allowing precise manipulation of experimental conditions.

1.2.1 Internal energy distribution of molecular beam

Effusive beams are characterized by Maxwell-Boltzmann distribution of velocity and internal energy whereas supersonic beams exhibit extensive cooling of translational, rotational and some relaxation of vibrational degrees of freedom. For a given nozzle temperature (T_N) the extent of internal energy cooling is determined by the stagnation pressure (P_0), nozzle diameter (d_N), and the collision number (Z). To quantify the rotational temperature

of the beam, we used rotational population intensity expression at a given temperature (T), defined as,

$$\frac{N_J}{N} = \frac{(2J+1)}{q_r} e^{-BJ(J+1)/kT}, q_r = \Sigma_J (2J+1) e^{-BJ(J+1)/kT} \tag{2}$$

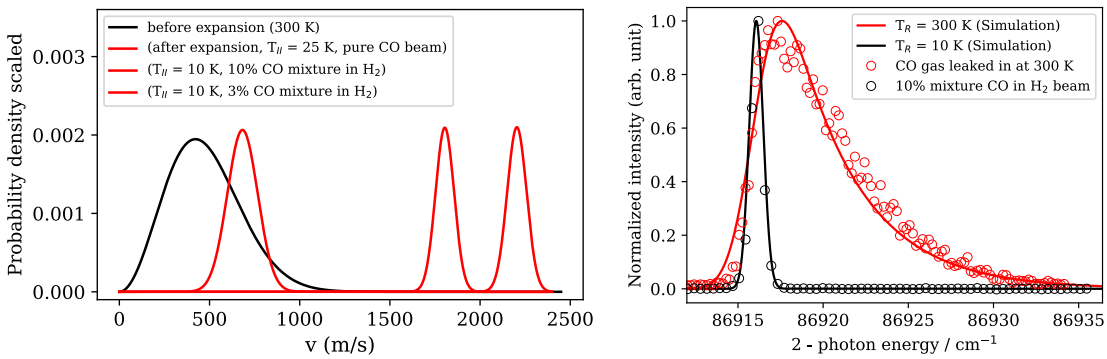

Figure 3: (Left) Velocity distribution calculated (solid black line) and measured (solid red lines) for CO before and after expansion from a reservoir at 300 K. The Maxwell-Boltzmann distribution of molecular speed in the reservoir is shown in black, and the measured velocity distribution and translational temperature after expansion of different mixtures are shown in red curves. (right) Black and red open circles represent REMPI spectra of seeded (10% CO in H_2) molecular beams and leak CO gas at 300 K in the UHV chamber. The black and red solid lines show the spectral simulation was done at 10 K and 300 K rotational temperature.

To fit the rotational distribution of the 0–0 band, we used the expression,

$$I(E) = \Sigma_J \Gamma(E - E_J) \frac{N_J}{N} \tag{3}$$

where I is the observed line intensity at the 2-photon energy E, $\Gamma(E - E_J)$ is the linewidth function (laser linewidth) for each rotational transition Q_{JJ} with energy E_J. Figure 3 shows 2+1 REMPI spectrum of the CO (0–0) Q_{JJ} band (black open circle) to a fit obtained function as given in equation 3. Fit function (black, solid line) corresponds to 10 K of rotational temperature. To verify our simulation function and method, we fit to thermal CO beam (CO gas leaked at 300 K in UHV chamber) that is in good agreement with experimental data as shown with a red solid line in Figure 3.

Our measurement confirms the successful preparation of a molecular beam with a maximum population at the lowest rotational states. The determination of translational energy and temperature, as outlined in Section 4, are additional crucial parameters characterizing the molecular beam. The scattering setup also features Helium reflectivity measurement, allowing real-time surface coverage monitoring and qualitative assessment of surface cleanliness through sputtering and annealing. A concise discussion of this tool is presented in the following section.

1.3 Helium reflectivity measurement for monitoring real-time surface coverage

In the context of reactive scattering, the initial sticking probability (S_0) serves as a key parameter expressing the reaction rate in molecular beam surface scattering experiments. Systematic measurement of S_0 across variables such as incident translational energy, angle, internal energy, orientation, surface temperature, and coverage is essential for unraveling the dynamics of surface reactions. It represents the fraction of incident molecules undergoing dissociative adsorption on the surface, providing fundamental insights into

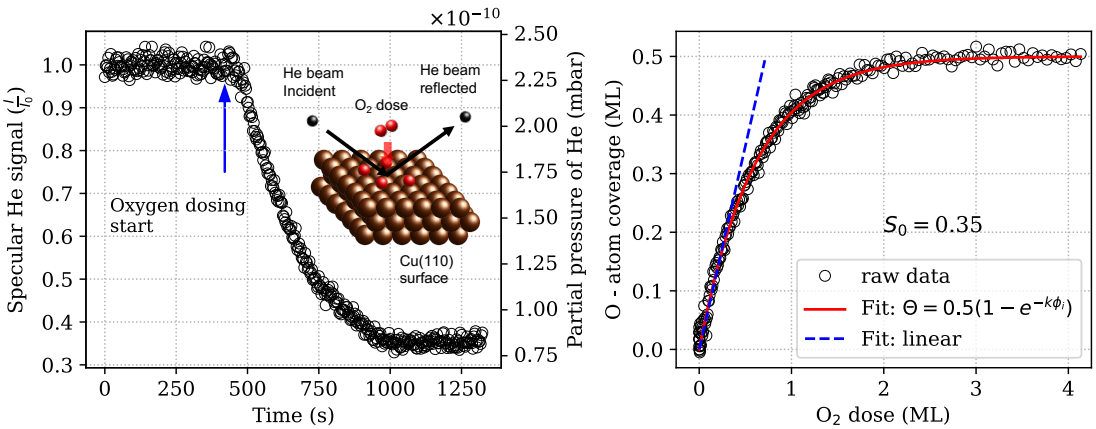

Figure 4: (left) Specular He signal (normalized) from the Cu(110) surface as a function of time, as the surface was exposed to oxygen (red arrow). The reflected He signal as observed by the mass spectrometer is also shown on the y-axis on the right. (b) Surface coverage of O-atoms as a function of oxygen exposure (in ML), obtained using the data shown in the left panel. The red curve shows a fit to a model assuming 0.5 ML as the saturation coverage. The blue line (dashed) shows a linear fit to the initial part of the curve. We estimate the initial sticking probability to be 0.35 (using linear fit)

reaction mechanisms. Different methods, such as AES, XPS, work function, and TPD, can determine adsorbate coverage, but Helium reflectivity stands out for its advantages. Helium atoms, with their soft probe nature and long-range interaction, provide high surface sensitivity (0.001 ML) and prevent chemical interactions with the surface. In essence, except for He reflectivity, other methods involve discrete measurements, introducing complexities such as extended measurement times. Helium reflectivity provides continuous measurements, offering high surface sensitivity (0.001 ML) and a unique ability to discern site-dependent reactivity. The adsorbate-induced attenuation of the specular He beam intensity can be utilized to monitor various surface coverage-related processes. In the case of random adsorption, the relative magnitude of the specular beam intensity (I/I_0) decreases exponentially with the coverage (Θ) and is defined as

$$I/I_0 = e^{(-\Sigma n_s \Theta)} \tag{4}$$

where I_0 is the specular He beam intensity for the clean surface, Σ is the total collision cross-section of adsorbed atom/molecule on the surface and n_s is the number of atoms per centimeter square. Figure 4 (left), it can be seen that as the oxygen dose increases, the relative magnitude of the He reflectivity signal decreases exponentially. Figure 4 (left) demonstrates the exponential decrease in He reflectivity signal with increasing oxygen dose, allowing estimation of S_0 (Figure 4, right) from the adsorbed O_2 flux. A comprehensive description and implementation details are provided in our previously published work [7].

Having covered essential steps like surface and molecular beam preparation, I now shift focus to our main objective: understanding CO_2 dissociation on metal surfaces.

2 Does CO_2 dissociate on Cu(110) surface ?

2.1 Background

Understanding the conversion of CO_2 to methanol is crucial for mitigating its environmental impact [8]. Due to CO_2's thermodynamic stability, its transformation into methanol presents significant challenges. Copper-based catalysts, utilizing H_2 and CO as co-reactants, are commonly employed in industrial processes for CO_2 conversion to

methanol. Previous research has identified CO_2 as the primary carbon source in methanol formation, with CO_2 dissociation on the catalyst surface being a key step in the overall reaction mechanism [9, 10]. Among low-index copper planes, Cu(110) surfaces have demonstrated promising catalytic activity by both experimental and computational studies .[11–14]. However, fundamental questions regarding dissociation probabilities and their energy dependence have remained unanswered, hindering a comprehensive understanding of the reaction mechanism. To address these questions, our study investigates the dissociation of CO_2 on clean Cu(110) surfaces under ultra-high vacuum (UHV) conditions. By measuring absolute dissociation probabilities (S_0) and analyzing their dependence on the incidence energy of CO_2, we aim to shed light on the microscopic details of the dissociation process. Our findings provide valuable insights into CO_2 dissociation on Cu surfaces, which are essential for the development of efficient catalytic processes aimed at mitigating CO_2 emissions. A detailed description of this work can also be found in our recently published study [15].

2.2 Methods

We examined the dissociation of CO_2 on Cu(110) at a surface temperature of 300 K using a recently developed molecular beam-surface scattering apparatus (see Figure 2). Previous studies have shown that CO desorbs from Cu(110) surfaces above 200 K, while O-atoms strongly bind to the surface, with the adsorbed layer remaining even at high temperatures up to 770 K [16]. Hence O- atom coverage build-up provides a direct measure of CO_2 dissociation. The O-atom coverage resulting from CO_2 dissociation was determined using Auger electron spectroscopy (AES) with a 1 μA surface current and 2.5 keV beam energy.

3 Results

3.1 Translational energy dependence of S_0:

Figure 5 shows surface O-atom coverage (in ML) as a function of incident CO_2 dose (in ML) for seven translational energies ranging from 0.098 eV to 1.59 eV, along with their best-fit curves. These measurements, conducted at $T_s = 300$ K and $\theta_i = 0°$, reveal a consistent pattern where all curves (except for $E_i = 0.098$ eV) converge to a saturation coverage of 0.5 ML. The initial slope increases with higher E_i, indicating translationally activated dissociation. For the lowest energy of 0.098 eV, even after dosing with 1200 ML of CO_2, the O-atom coverage remained indistinguishable from the background, suggesting a S_0 below detection sensitivity. In Figure 6 (left), S_0 values derived from initial slopes are plotted against translational energy, with black circles representing measurements at $\theta_i = 0°$ and a red triangle denoting $\theta_i = 19°$ ($E_i = 1.59$ eV, $E_n = 1.42$ eV). An empirical S-shaped curve (discussed below) fits the data, showing S_0 increasing from 3.9×10^{-4} to 1.8×10^{-2} with increasing E_n from 0.64 to 1.59 eV.

3.1.1 Barrier height estimation using 1-D potential energy model

The S_0 obtained for the measurement at $\theta_i = 19°$ shows that only E_n is relevant for overcoming the dissociation barrier, indicating that a simple one-dimensional barrier model can be used to understand this system. In this case, the overall sticking probability can be expressed as:

$$S_0(E,T) = \sum_v F_B(v,T) \cdot S_0(v) \tag{5}$$

Where $F_B(v,T)$ is the population in different vibrational states at a given vibrational temperature (T) of the incident beam, and $S_0(v)$ is the vibration state-specific initial

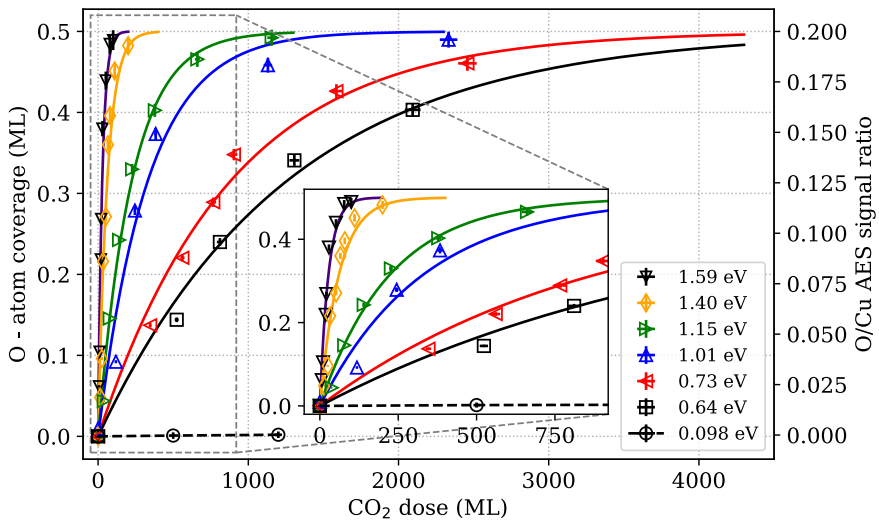

Figure 5: A combined plot depicting O-atom coverage build-up (in ML) on Cu(110) surface as a function of incident CO_2 dose (in ML), measured for different translational energies. The inset shows a zoomed-in view of the initial kinetics. At the lowest energy, we could not observe any O-atom coverage build-up. Clearly, the S_0 increases with incident translational energy indicating a translationally activated dissociation.

dissociation probability. Given that in our experiments, the nozzle is at room temperature (300 K), to a good approximation $F_{\mathrm{B}}(v = 0, T) = 1$, i.e., the population from the higher vibration states is much smaller than the ground state. The expression for $S_0(v)$ is given

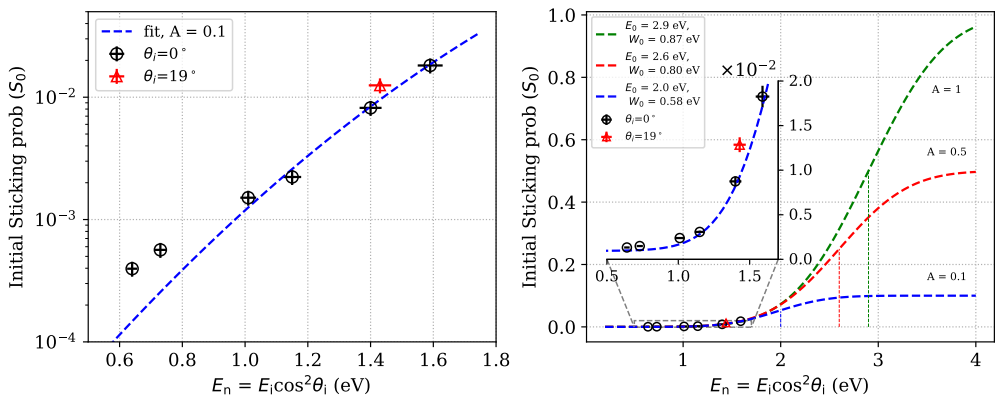

Figure 6: (left) A plot of S_0 obtained at different E_{n} of CO_2. Black circles show the measurements at normal incidence, and the red triangle corresponds to a measurement at $\theta_i = 19°$. (right) The same points (on a linear scale) are shown along with the best fits using S-shaped curves with different values of A of 0.1 (blue), 0.5 (red), and 1 (green). Even for the lowest A (0.1), the E_0 comes out to quite large and is 2.0 eV.

by a S-shaped curve (equation 6). Here, the saturation value of S_0 is given by A, E_0 corresponds to the average value of the dissociation barrier (also the midpoint of the curve), and W_0 describes the width of the distribution of barrier heights [17–19].

$$S_0(v) = \frac{A}{2}\left\{1 + erf\left[\frac{E_{\mathrm{n}} - E_0}{W_0}\right]\right\} \tag{6}$$

In Figure 6, even at the highest incidence energy (1.59 eV) used, the S_0 remains low (1.8×10^{-2}) and is still increasing. With these results only an approximate estimation of the best-fit parameters such as E_0 and W_0 is feasible. With the S_0 continuing to rise even

at the highest E_n used, a realistic estimate for the maximum S_0 (A) would be around 0.1. Consequently, the resulting E_0 is 2.0 eV with $W_0 = 0.58$ eV (from the fits shown in the blue dashed curve, Figure 6, right).

It is worth pointing out here that according to previous work [20], at $E_n = 1.3$ eV, using a heated nozzle at 750 K, no dissociative chemisorption was observed. Even more striking is the fact that the dissociation barrier estimated from our observations is much larger compared to 0.64 eV reported previously [11, 14]. In the section 3.2, we discuss a few hypotheses based on which these seemingly large differences can be rationalized.

3.1.2 Direct measurement of S_0 using molecular beam reflection method

We used the molecular beam reflection method (King and Wells) to determine the sticking probability during CO_2 dissociation [21]. By measuring changes in CO(g) concentration or CO_2 reduction, we assess dissociation probability without requiring additional calibrations such as flux, AES, or pumping speed. While implementation may be challenging for dissociation probabilities below 5%, measuring the CO to CO_2 ratio offers better sensitivity, enabling the detection of reaction probabilities well below this threshold.

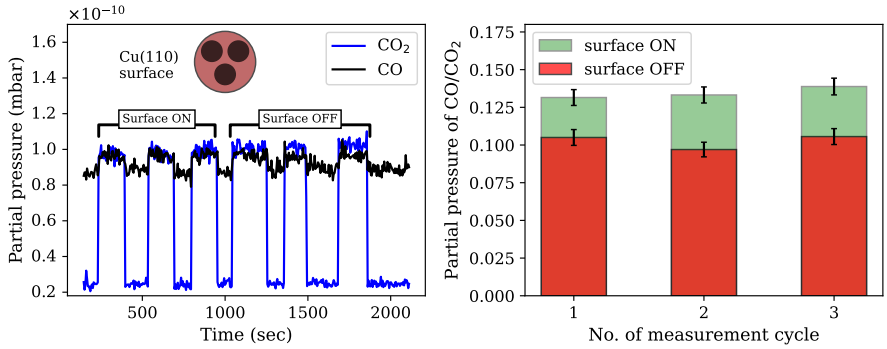

Figure 7: The left panel illustrates multiple cycles of CO_2 dosing with and without the surface recorded in QMS. A visual representation of the Cu(110) surface, along with three different spots used to dose CO_2 flux, is shown in the left panel. (right panel) displays the plotted ratio of CO/CO_2 for the same. Measurement uncertainty is associated with determining the mean position of the distribution.

These measurements utilized a 1.65 eV CO_2 beam dosed on a clean Cu(110) surface (see Figure 7, left). Comparisons were made with and without the surface (surface removed from the line of sight of the incident CO_2 beam), measuring the CO to CO_2 ratio in QMS. Without the surface, a 10% ratio, attributed to electron impact fragmentation of CO_2, was observed. With the surface present, three measurements at different locations showed a ratio increase of 2.5% ± 0.5%, corresponding to a sticking probability (S_0) of 1.9% ± 0.3%.(see Figure 7, right). This aligns closely with reported values from O-atom coverage methods (see Figure 6).

3.2 Discussions

3.2.1 Comparison with previous molecular beam studies

In contrast to our finding, Funk and co-workers [20], who employed molecular beams under UHV conditions to investigate CO_2 physisorption dynamics on Cu(110), reported that no dissociative chemisorption of CO_2 was observed. At a cold surface below CO_2 desorption temperature (90 K), they reported a S_0 of 0.05 ($E_n = 1.3$ eV) for non-dissociative physisorption, approximately ten times higher than our observed S_0 for dissociative chemisorption at the same incidence energy (see Figure 6, left). This discrepancy likely arises from rapid buildup of physisorbed CO_2 coverage at low T_s, leaving few sites

available for dissociation. Our simple kinetics-based model, incorporating S_0 values from Funk et al. [20] and our work, supports this interpretation.

3.2.2 Dissociation barrier Threshold: Comparing our findings with high-pressure and DFT studies

Based on previous high-pressure measurements [11], revealing dissociation probabilities ranging from 1×10^{-11} to 1×10^{-9} at pressures of 65 and 650 Torr, with a reported activation barrier of 0.64 eV. Recent ambient pressure X-ray photo-electron spectroscopy studies [14] corroborate these findings, reporting a dissociation probability of 4.4×10^{-11} per collision. However, our observations show an increasing trend in S_0 even at much higher E_i than 0.64 eV, contrary to approaching the reported activation barrier, suggesting other contributing factors.

Differences in surface structures under low vs. high-pressure conditions may influence dissociation barriers, despite similar O-atom coverages observed in both studies. Therefore, the significant deviation in dissociation barriers likely arises from other factors. Another point to be considered is that distinct reaction pathways are expected under high-pressure conditions, favoring a precursor-mediated pathway compared to molecular beam experiments and subsequently following a low activation barrier pathway for dissociation. Given the large difference in the estimated barriers in the case of CO_2/Cu(110), it will be very interesting to look into the underlying potential energy surface and trajectory calculations to understand the reaction pathways. Finally, large differences in activation barriers can also arise from different reaction sites, i.e., steps vs. terraces in high vs. low-pressure studies. Similar findings were reported for N_2 dissociation on Ru(0001) [22, 23], with a 0.4 eV barrier under high pressure versus > 1 eV in molecular beam studies [24]. Blocking step sites led to a significant reactivity decrease (factor of 10^9) under high pressure, but a much smaller reduction (2x) in molecular beam experiments. This suggests a preference for step sites under high pressure and terrace sites in molecular beam experiments.

In summary, our molecular beam experiments reveal a substantial dissociation barrier for CO_2 on Cu(110) terrace sites, estimated to be around 2 eV. Comparison with high-pressure studies suggests that step sites could significantly lower the activation barrier under those conditions, contrasting with terrace sites observed in molecular beam experiments. This highlights the need for a reevaluation of the microscopic understanding of CO_2 dissociation on Cu surfaces. Additionally, exploring the reactivity of vibrationally excited CO_2 could provide further insights into the dissociation barrier.

In our investigation of CO_2 reaction dynamics on a Cu(110) surface, we estimate the barrier height using the assumption of saturation values (A=0.1, 0.5, and 1). Achieving saturation requires a higher kinetic energy beam, which can be generated using a hot nozzle. However, this introduces complexity, and the distribution of internal energy (translational, rotational, and vibrational) must be measured for quantitative conclusions. Another key focus is understanding collision dynamics, particularly non-reactive scattering, of CO_2 on the same surface. I've developed a tool for state-specific scattered flux detection alongside a hot nozzle. Further details are discussed in the following section.

4 Design and characterization of ion imaging setup for understanding dynamics of non-reactive scattering

4.1 Background

Understanding non-reactive scattering requires knowledge of state-specific information before and after the scattering event. Imaging techniques combined with a laser-based resonant ionization (REMPI) scheme, capable of measuring both internal energy and velocity

distribution were first developed by Chandler and Houston [25]. These methods enable the measurement of both internal energy and velocity distribution. Velocity-Map Imaging (VMI) technique, introduced by Eppink and Parker, offers a significant resolution improvement, mapping ion velocities to different detector locations using an Einzel lens [26]. While initially demonstrated in cross-molecular beam experiments, its implementation in surface scattering experiments was demonstrated by Harding et al. [27, 28]. The advantages of ion imaging in measuring complete 2D velocity distribution along the scattering plane in a single measurement.

The upcoming section will detail ion optics design, optimization, experimental setup, testing, calibration, and its application in surface scattering experiments. Characterization involves measuring incident and scattered flux velocities of both CO and CO_2 on a Cu(110) surface.

4.2 Ion optics and Experimental setup

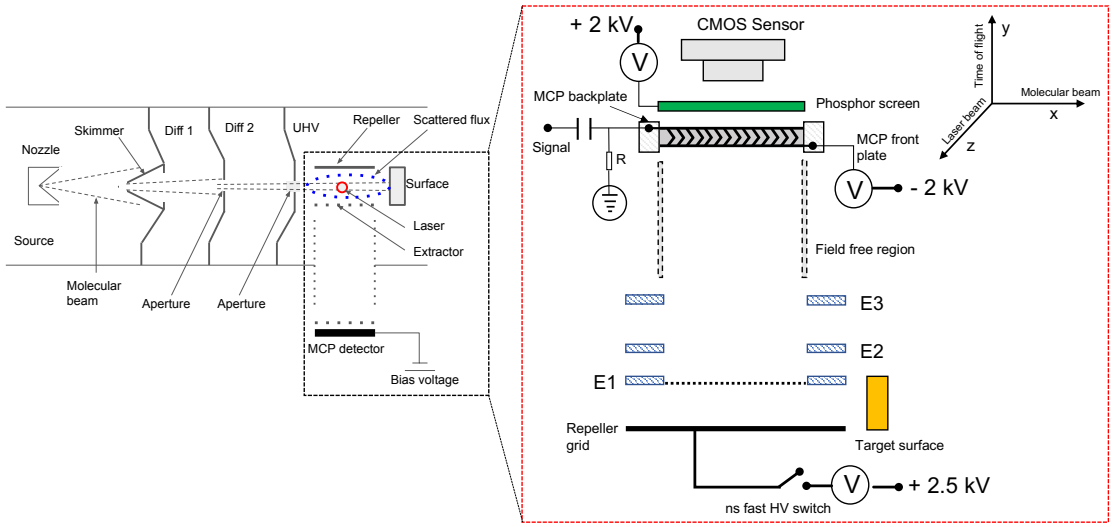

Figure 8: (left) shows a schematic of surface scattering apparatus along with ion imaging setup. (right) A zoomed detailed view of the schematic drawing of the imaging setup. The gap between the repeller grid and E1 grid is 8 mm. Total flight length is 280 mm.

Figure 8 provides a schematic representation of our experimental apparatus, designed to be compatible with the surface scattering beam apparatus. The ion optics design allows us to operate in both spatial imaging and Velocity Map Imaging (SVMI) modes, following ideas discussed in literature [27, 28]. Spatial mode is currently being used. In the spatial imaging configuration, the ion optics consist of a repeller and a grounded flat grid electrode, producing a uniform electric field that maps ion positions onto the detector. A dual-stack of multichannel plates (MCPs) and a phosphor screen are used for ion detection, with MCPs and the phosphor screen biased at -2 kV and +2 kV voltage, respectively. We employed a 2 + 1 REMPI scheme to ionize molecules like CO and CO_2. The Rayleigh length of the laser beam allows ionization across the full extraction region, enabling the detection of the full in-plane scattering distribution. Velocity information is obtained by considering the time of flight of ions from the laser ionization position to the detector, laser ionization position, and spatial calibration of images. Using this information, a single image can be converted into a speed distribution, which can be fit by a function.

$$f(v) = v^3 exp[-\frac{(v - v_0)^2}{\alpha^2}] \tag{7}$$

One method involves fitting $f(v)$, the flux as a function of velocity v, to determine the

mean energy and width of the energy distribution. Here, v_0 represents the flow velocity, and $\alpha^2 = \frac{2k_B T_p}{m}$, where T_p is the characteristic temperature in the frame of reference moving with v_0. Another approach is to plot the most probable positions as a function of delay time, resulting in a straight line whose slope represents the most probable speed of the molecules (Figure 11). This method offers the advantage of not requiring knowledge of either the position of the laser or the ion flight time.

4.3 Testing and calibration of imaging field

To ensure the proper operation of an ion imaging setup, maintaining a uniform electric field between the repeller and ground electrode (extractor region) is essential. Challenges such as distortions from ion optic edges and maintaining the flatness of the mesh and grounded target surface arise. Systematically scanning laser ionization positions throughout the extractor region, along with concurrent measurements of ion images at the MCP, revealed a consistently uniform electric field within a 14 mm region, even in the presence of a surface as shown by the red square dashed box (Figure 9, right).

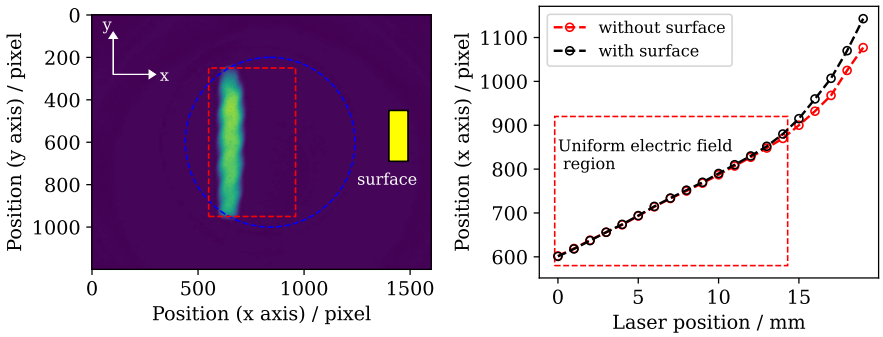

Figure 9: In the (left) panel, the ion image from leaked CO gas at 300 K exhibits two features: the Rayleigh length of the laser surpasses the 40 mm extractor region (shown in dashed blue circle). The red dashed square box outlines the electric lines that maintain uniformity across the entire region, and the rectangular yellow patch denotes the target surface. Note that the laser beam is along the y-axis, the molecular beam is along the x-axis, and the TOF is along the z-axis. In the (right) panel, the mean ion image position is plotted by varying the ionization laser position throughout the extractor region. Proximity to the target surface and the electrode edge leads to non-uniform electric lines.

4.4 Incident and scattered speed distribution measurements

In our imaging setup, I systematically measured the incident beam energy of a CO molecular beam by varying the kinetic energy using gas mixtures of 3%, 10%, and 100% CO in H_2. The corresponding calculated kinetic energies were 0.8 eV, 0.5 eV, and 0.082 eV, respectively, as determined by equation 1. The measured kinetic energy for each mixture was consistent with the calculated values (see Table 1).

I conducted CO_2 scattering experiments on a Cu(110) surface, simultaneously measuring incident and scattered CO_2 flux. Employing a 2+1 REMPI CO detection method generated from the photo-dissociation channel of CO_2. Typical images (Figure 10, left) show an intense incoming molecular beam and a broad line representing scattered molecules. Movement induced by pulsed extractions causes shifts in both beams. For data analysis, plotting the most probable positions against delay time provides molecule speeds (Figure 11), aiding ionization position and time-of-flight determination.

I have shown results for this ion imaging setup capable of directly imaging the scattering of molecules from a clean surface that will be used to understand non-reactive scattering. Further developments in future will aim to test the instrument in SVMI mode

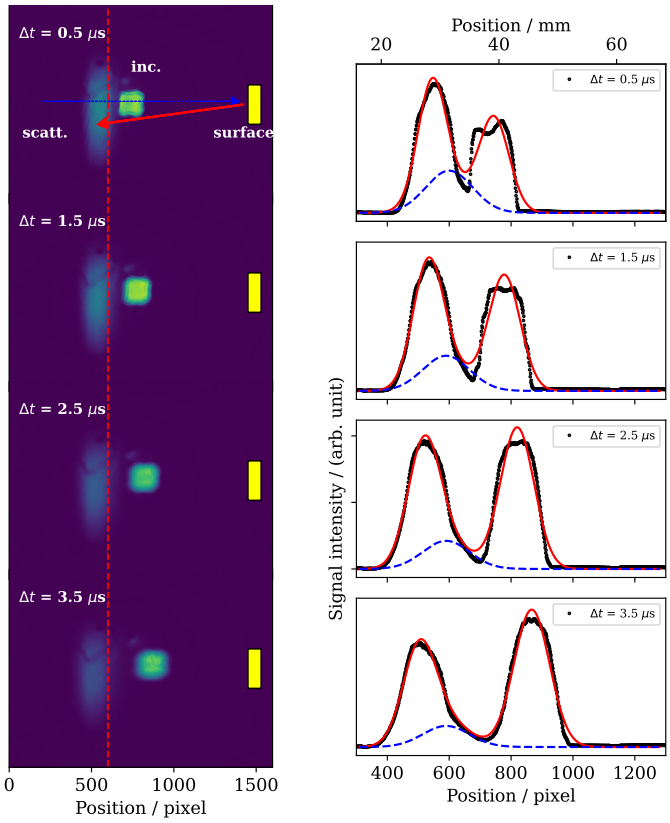

Figure 10: (left) Ion images for CO_2 scattering from Cu(110) surface. The nominal extraction delays are shown. The labeled features in the top image are the incident (inc.) and scattered (scatt.). The position of the surface is shown by the yellow rectangle, and the incident beam by the blue arrow. (right) Ion position distributions for different extraction delay times. The movement of the ion peak can be seen. The black points show experimental data, and the red lines are the fits used to determine the center positions. A third feature is shown in blue dotted which corresponds to background buildup

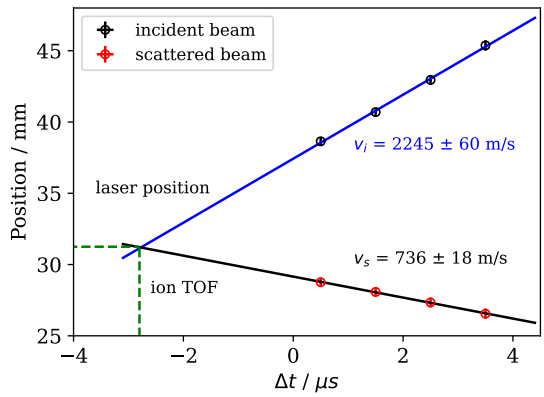

Figure 11: This Panel shows the peak center positions as a function of extraction delay time. The most probable speed of the ions is obtained from the slope of a straight line fit (shown in blue) for the incident and (shown in black solid line) for scattered to these positions. The error bars show an estimate of the uncertainty in the fit to the center of the scattered ion peak.

for improved speed resolution. This tool also allows us to characterize a molecular beam produced from the hot nozzle, extending our study of CO_2 dissociation at higher kinetic energies. By measuring internal energy, we can distinguish the effects of vibrational and translational energy in dissociation dynamics.

5 Ongoing work: Understanding of the role of vibrational energy in CO_2 dissociation dynamics on Cu(110) surface

In our previous work, I investigated the translational energy of CO_2 in its dissociation dynamics. Now, we turn our attention to understanding the impact of vibrational energy on CO_2 reaction dynamics, discerning whether the dissociation barrier is early (favoring translational energy) or late (favoring vibrational energy). Notably, at a 300 K nozzle temperature, I achieved a maximum 1.6 eV translational energy for the CO_2 beam, corresponding to a 2% reaction probability, still below saturation (see Figure 6). To increase kinetic energy, I will use a heated nozzle (up to 900 K), yielding up to 3 eV kinetic energy and a 20% population in the excited vibrational state (bending state) at 900 K nozzle temperature. Our ongoing work focuses on characterizing the hot nozzle to extend ground state CO_2 dissociation studies and explore the role of vibrational energy in dissociation processes.

Table 1: Experimental measured (E_{meas}) and calculated (E_{cal}) translational energies (in eV) and rotational temperature (in K) of 6% CO mixture in H_2

Nozzle temperature (K)	E_{meas} (eV)	Rotational temperature (K)	E_{cal} (eV)
300	0.57	30	0.62
450	0.78	30	0.82
723	1.24	35	1.30
807	1.41	45	1.45

Energy measurements unequivocally demonstrate the capability to achieve a more than twofold increase in translational energy by varying the nozzle temperature from 300 K to 800 K, with an associated uncertainty in E_{meas} of $\pm 5\%$. We expect that, under identical expansion conditions, the CO_2 beam will manifest similar characteristics, enabling the production of a CO_2 beam with energies beyond 3 eV.

The proposed approach will be used to understand the role of vibrational energy in dissociation dynamics. However, it lacks specificity regarding individual vibrational modes' reactivity due to the statistical distribution of vibrational modes at high nozzle temperatures. For mode-specific dissociation studies, state-specific preparation incident reactants are needed which need a narrow linewidth radiation source in the IR region such as OPO.[29]. I have developed a wavelength meter capable of actively stabilizing the IR radiation source's wavelength, enabling these studies. Further details on this tool are discussed in the forthcoming section.

6 A novel approach to build high accuracy and precision wavemeter

6.1 Background

Accurate wavelength measurement is essential for various applications such as atomic and molecular spectroscopy, laser frequency characterization, and metrology. For example, in tasks like surface scattering of excited molecules, where excitation profiles range from a few MHz (due to transit time broadening) to a few 100 MHz (due to power broadening from pulsed lasers), precise identification and locking of transitions are crucial, often necessi-

tating wavemeters with accuracy levels of 1 part in 1×10^6 and precision better than 30 parts in 1×10^8. Techniques like the Michelson and Fizeau methods are commonly used for achieving high precision and accuracy in both pulse and continuous lasers. Snyder introduced the Fizeau-based wavemeter, offering advantages over Michelson, including a static design, insensitivity to intensity fluctuations, and suitability for low repetition rate pulsed lasers [30]. However, Snyder's design required vacuum conditions and faced dispersion errors due to vacuum enclosure windows. Efforts have been made to address these challenges and enhance accuracy [31–34]. Lee and co-workers developed multi-wedge designs for precise measurements (up to 2.5 parts in 1×10^6), but constructing them proved challenging and costly.

To address these issues, I have worked on the design, development, and characterization of a cost-effective approach using equilateral prism stacks and solid wedges which is referred to as cross fringe Fizeau interferometer (CFFI). This setup ensures orthogonal wedge axes and enables fringe pattern capture using an affordable webcam sensor.

6.2 Experimental design and method

A monochromatic, collimated beam incident on two uncoated optical flat surfaces inclined at a small wedge angle generates straight, parallel fringes. The fringe intensity, $I(x)$ is described by:

$$I(x) = \frac{I_0}{2}(1 + Cos[\frac{2\pi(x - x_0)}{\Gamma}]) \tag{8}$$

Here, $I(x)$ represents the intensity at position x, I_0 is the maximum intensity, x_0 is the position of the maximum intensity, and Γ denotes the fringe spacing. An initial estimate of the wavelength (first pass wavelength, λ) can be made using the fringe spacing (Γ) and wedge angle (α):

$$\lambda = 2\Gamma tan\alpha \tag{9}$$

where λ is the first pass wavelength (FP). The first-pass calculated wavelength may not provide sufficient accuracy due to various factors such as the flatness of the optics, the homogeneity of the sensor, and any potential sources of noise or interference.

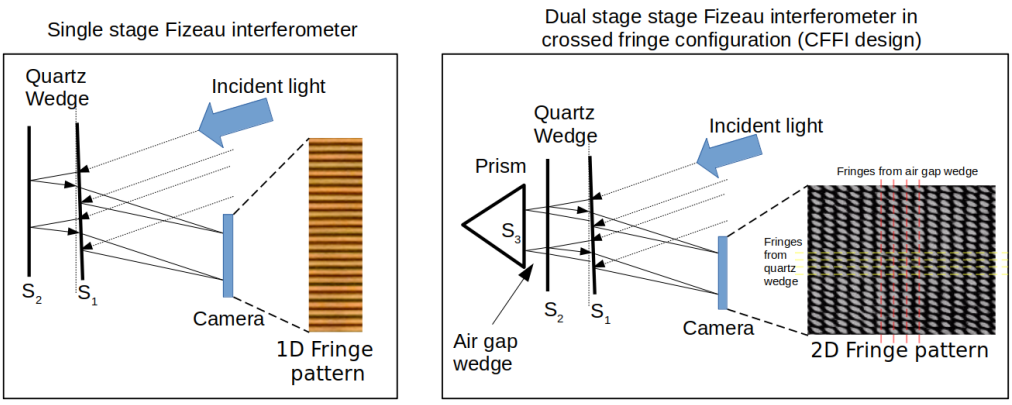

Figure 12: (Left) A schematic diagram illustrating the working principle of a single-stage Fizeau interferometer. The partial reflections of the incident light on the front and back surface (S1 and S2) of a quartz wedge interfere and give rise to an interference pattern consisting of a series of evenly spaced bright and dark fringes (1D fringe pattern). (Right) Working principle of the CFFI design with a 2-stage Fizeau interferometer. Here an additional surface S3 (face of an equilateral prism) is placed behind the quartz wedge creating a second wedge (air gap wedge). The quartz and the air gap wedge are oriented orthogonal to each other and the resulting interference pattern resembles a grid of evenly spaced bright spots (2D fringe pattern). This 2D pattern is a product of the two orthogonal 1D patterns generated from the quartz (parallel to dashed yellow lines) and air gap wedge (parallel to the dashed red lines). The wavelength of the incident light is estimated by analyzing the 2D interferogram captured by a camera

A more refined wavelength estimate (second-pass wavelength) can be made by knowing the correct order number and the optical path length provided that the uncertainty of the FP wavelength is within the free spectral range (FSR) of the optical wedge, typically expressed as $\frac{\delta\Gamma}{\Gamma} < \pm\frac{FSR}{2}$. The second-pass wavelength (λ_s) is calculated by:

$$\lambda_s = \frac{2(e_s + x\tan\alpha)}{2m_s + 1} \tag{10}$$

Here, e_s is the wedge spacing, m_s is the order number, and λ_s is the second-pass wavelength.

6.3 Accuracy and precision of CFFI

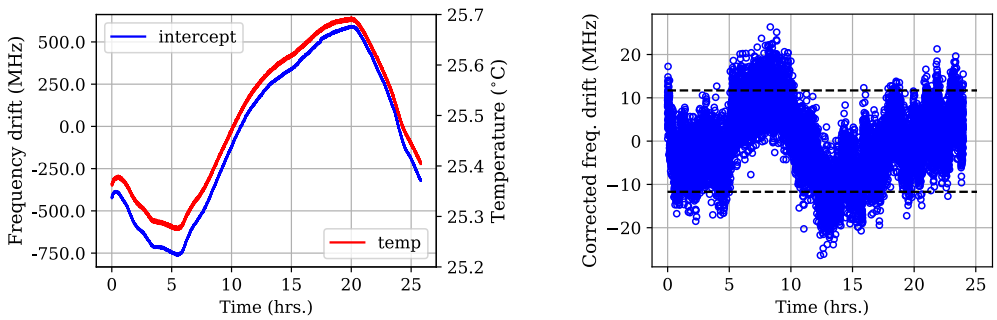

Figure 13: Performance of a thermal drift corrected wavemeter. (right) The temperature-corrected frequency shows a deviation of about 25 MHz (3σ, shown with two horizontal dashed lines) over a 24-hour timescale. The light used here in the source is a ppb-level precision frequency stabilized red He Ne laser.

Utilizing a home-built polarization-stabilized He-Ne laser (stability, $3\sigma = 0.42$ MHz, 18 hours) for characterizing the performance of CFFI [35]. Attempts at achieving active stability in the MHz range by thermally stabilizing the entire wavemeter box faced challenges due to its slow thermal response. However, an alternative approach involved using a high-precision temperature sensor with ± 0.1 °C accuracy and 0.002 °C precision. This method successfully compensated temeperature dependent frequency drift in the wavemerter up to a level of 25 MHz over 25 hours. (see Figure 13).

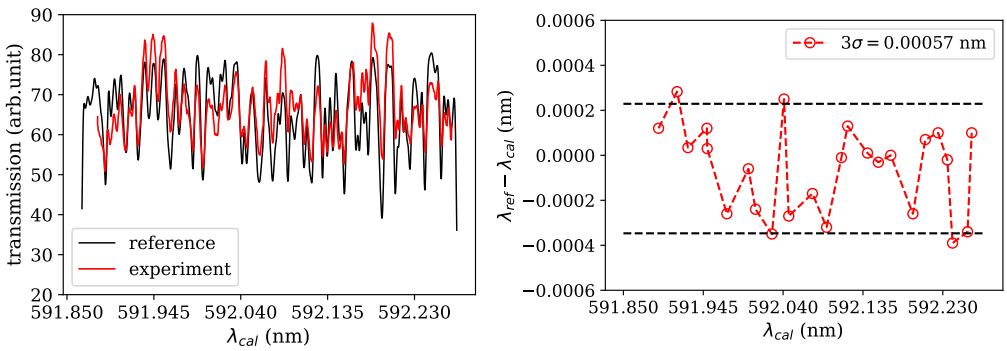

Figure 14: (top) Measurement of the iodine spectrum using a tunable dye laser. The values on the x-axis correspond to the wavelengths measured using the wavemeter (λ_{cal}). A clear one-to-one overlap in the expected and measured peak positions is observed. (bottom) Deviation among the wavelengths measured using the wavemeter and that from the known spectral lines for Iodine absorption. The overall deviations are quite small and at the sub-ppm level.

The CFFI wavemeter calibration within the 100 nm range employed two pulse dye lasers: SIRAH (570 nm to 608 nm) and Lioptec (634 nm to 670 nm), achieving an accuracy

of 0.005 nm. This accuracy was primarily limited by non-linearity in the calibration source. However, we achieved excellent accuracy by measuring the iodine spectrum using calculated wavelengths from the second pass. The achieved accuracy was sub-parts per million (ppm), considering three standard deviations (see Figure 14). Notably, limitation to accuracy was a result of severeal factors such as the Doppler linewidth (1-2 GHz) of the iodine lines, the laser linewidth (1.5-2 GHz), passive stability and multimode nature of the dye laser.

In summary, our CFFI setup demonstrates exceptional performance, achieving ppm accuracy (1 part in 1×10^6) in the 100 nm wavelength range and 25 MHz frequency deviation for 24 hours (3σ). This precision is crucial for high-resolution spectroscopy, and we regularly use it for quantum state-specific studies in our lab, including identifying IR transitions in molecular beams and simultaneous locking. The multistage design with off-the-shelf optics, along with low-cost 2D sensors, makes it a simple, robust, and reliable tool.

7 Structure of the thesis

Chapter 1: Introduction

Background on CO_2 surface chemistry, existing studies, and literature survey for molecular beam surface scattering studies.

Chapter 2: Experimental setup for surface scattering studies

UHV experimental setup, Preparation and characterization of the molecular beams, surface preparation for scattering experiments.

Chapter 3: Understanding CO_2 dissociation dynamics on Cu(110)

Background, Experimental setup, Quantification of CO_2 dissociation, O-atom uptake measurement with CO_2 dose, Flux estimation, Incident energy dependence S_0.

Chapter 4: Understanding of role of vibrational energy in CO_2 dissociation dynamics on Cu(110)

Hot nozzle testing and characterization, Quantifying CO and H_2O formed due to RWGS reaction, Vibrational efficacy, and barrier nature (late or early barrier) estimation

Chapter 5: Design and characterization of Ion imaging tools for non-reactive scattering studies

Background, ion optics design optimization using SIMION, calibration, characterization

Chapter 6: Design, development, and characterization of CFFI-based wavemeter

Details of the development of the home-build wavemeter design, working principles, operating procedure, and application in quantum state-specific surface scattering studies.

Chapter 7: Future directions and concluding remarks

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Research Publications

Accepted/published:

1. **Saurabh Kumar Singh**, and Pranav R. Shirhatti. The curious case of CO_2 dissociation on Cu(110). *The Journal of Chemical Physics* **160.2** (2024).

2. **Saurabh Kumar Singh**, Avinash Kumar, and Pranav R. Shirhatti. A simple and low-cost setup for part per billion level frequency stabilization and characterization of red He-Ne laser. *HardwareX* **14 (2023): e00421**.

3. Bhardwaj, Geetika, **Saurabh Kumar Singh**, and Pranav R. Shirhatti. A compact and highly collimated atomic/molecular beam source. *Rev. Sci. Instrum.* **94.4 (2023)**.

4. Avinash Kumar, **Saurabh Kumar Singh**, Pranav R. Shirhatti; A versatile and narrow linewidth infrared radiation source for ro-vibration state-selected preparation of molecules in molecular beams. *Rev. Sci. Instrum.* **94 (11)** (2023).

Patents:

1. **Saurabh Kumar Singh**, Avinash Kumar, and Pranav R. Shirhatti. A multi-stage thermal drift compensated cross-fringed Fizeau interferometer wavemeter and method thereof. **Patent:** Indian App. No. 202321048766 dated 20-Jul-2023 – TIFR

Manuscripts under preparation:

1. **Saurabh Kumar Singh**, Avinash Kumar and Pranav R. Shirhatti, Design and characterization of a high-performance wavemeter based on crossed fringe Fizeau interferometer

2. Amaraja Taur, **Saurabh Kumar Singh**, Pranav R. Shirhatti, Identifying signatures of thermal and non-thermal reaction pathways in plasmon-induced H_2 + D_2 exchange reaction. ChemRxiv. 2021; **doi:10.26434/chemrxiv-2021-87t1c.**

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